



ORIGINAL RESEARCH ARTICLE

Investigating the Roles of Horizontal Solidification and Heat Treatment Process Parameters in 6201 Alloy: Thermal, Microstructure, Electrical, Mechanical, and Cold-Rolling Analysis

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This study investigates the combined influence of horizontal directional solidification and subsequent T6 heat treatment on the microstructural evolution and the resulting electrical and mechanical properties of an Al-6201 alloy. Unsteady-state horizontal solidification was performed using a water-cooled mold system, enabling real-time thermal mapping during solidification. The acquired thermal data were used to determine the solidification parameters, including growth rate (V_L), cooling rate (T_R), and local solidification time (t_{SL}). Following solidification, the alloy was subjected to a T6 heat treatment consisting of solution at 530 °C for 3 h, water quenching at room temperature, and artificial aging at 200 °C for 2.5 h. Variations in T_R promoted a macrostructural transition from columnar to equiaxed grains and refinement of the Al-rich dendritic matrix. Finer secondary dendrite arm spacing (λ_2) contributed to increased hardness by hindering dislocation mobility. After the T6 treatment, precipitation hardening improved mechanical performance, resulting in a 51% increase in hardness and a 65% rise in ultimate tensile strength. The processed alloy achieved a tensile strength of 320.48 MPa, an electrical resistivity of 33.6 nΩ m, and an electrical conductivity of 51.2% IACS, evidencing a superior balance between strength and conductivity.

Keywords AlMgSi alloys, hardness, microstructure, rolling, unsteady-state horizontal solidification process

1. Introduction

Metals are fundamental to modern industry, with iron, aluminum, and copper being the most widely consumed, in that order. Aluminum, particularly in the form of alloys, has gained increasing importance in the transportation sector, including automotive, marine, and aerospace applications, mainly due to advancements in its mechanical performance. The low density of aluminum alloys contributes to reduced vehicle weight, improving fuel efficiency and consequently lowering environmental impact (Ref 1, 2).

In addition to their favorable strength-to-weight ratio, aluminum alloys exhibit excellent mechanical properties when combined with appropriate alloying elements. Aluminum has also become the second most used metal in electrical power transmission, owing to its high conductivity and low density. Among the various alloy systems, those containing silicon (Si) and magnesium (Mg) are of relevance, as their addition significantly enhances overall performance. Magnesium increases mechanical strength, ductility, and machinability, while also improving the balance between mechanical and corrosion resistance. Silicon improves melt fluidity, reduces volumetric shrinkage and the thermal expansion coefficient, enhances castability and weldability, and increases resistance to wear and corrosion (Ref 3-5).

In heat-treated Al-Si-Mg alloys, magnesium strengthens the Al-rich matrix (α -Al) through precipitation hardening associated with the formation of the β -Mg₂Si phase. During the solution and aging stages of the T6 heat treatment, appropriate control of this process promotes a uniform dispersion of fine β -

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Mg₂Si precipitates within the α -Al matrix, resulting in enhanced strength and hardness (Ref 5, 6). Precise control of the solution temperature is crucial, as excessively high temperatures may cause partial melting of eutectic structures, leading to the deterioration of mechanical properties, whereas insufficient temperatures can result in incomplete dissolution of secondary phases (Ref 7, 8).

The hardening mechanism follows a well-established precipitation sequence. After solution treatment and quenching, aging at relatively low temperatures promotes the nucleation of magnesium- and silicon-rich solute clusters, known as Guinier–Preston (GP) zones. These metastable clusters subsequently evolve into intermediate β'' and β' precipitates before transforming into the stable equilibrium β -Mg₂Si phase. The GP zones typically appear as needle-like precipitates approximately 10 nm in length, followed by homogeneous precipitation of β'' -Mg₂Si, heterogeneous precipitation of β' -Mg₂Si, and the eventual formation of the equilibrium β -Mg₂Si phase (Ref 4, 9).

Aluminum alloys of the 6xxx series, which contain Mg and Si as their principal alloying elements, have been extensively studied over recent decades. Among them, heat-treatable alloys such as 6101 and 6201 are considered the most promising for industrial applications due to their excellent combination of formability and mechanical strength (Ref 10, 11). Studies have shown that the 6201 aluminum alloy, after appropriate heat treatment, can exhibit high electrical conductivity together with superior mechanical strength (Ref 12, 13). Under these conditions, 6201 alloy wires are widely employed in electrical power transmission and distribution systems, particularly in the manufacture of AAAC (All-Aluminum Alloy Conductor) cables (Ref 14). AA6201 is a heat-treatable Al-Mg-Si alloy belonging to the 6xxx series, widely used in electrical conductors due to its favorable balance between mechanical strength and electrical conductivity.

Reports indicate that achieving an optimal balance between electrical conductivity and tensile strength in such wires is possible through the combined application of heat treatment processes—namely solutionizing and aging—along with subsequent cold working or plastic deformation stages (Ref 14). This combination promotes microstructural refinement and controlled precipitation, thereby enhancing both electrical and mechanical performance.

The influence of silicon content on electrical conductivity and hardness in 6xxx series Al-Mg-Si conductive aluminum alloys has been investigated for different aging times. Studies have also examined the effects of natural aging, as well as its combination with pre-aging, on the strength and electrical conductivity (τ) of cold-drawn Al-Mg-Si AA6201 conductive alloys. The results indicated that the combined application of natural aging and pre-aging, followed by cold drawing and post-aging, yielded the highest tensile strength (369 MPa) while maintaining an acceptable electrical conductivity of 53% IACS (Ref 12, 13).

It is well established that the physical and chemical properties of as-cast metals depend on the chemical composition, size, morphology, and distribution of the microstructural constituents. One of the most effective ways to control the structure of a metallic material is through the regulation of thermal parameters during solidification, such as the growth rate (V_L) and cooling rate (T_R) (Ref 15, 16).

Among the various physical properties, numerous studies have examined the influence of solidification structure on the

mechanical and electrical behavior of aluminum alloys (Ref 17–20). This has stimulated increasing interest in investigating different aluminum alloy systems produced by upward directional solidification (Ref 18, 20, 21), extending beyond the conventional 6xxx series. Additionally, the influence of silicon content on the electrical conductivity and hardness of 6xxx series Al-Mg-Si conductive alloys has been investigated under different aging conditions (Ref 13).

In conventional heat treatment of high-conductivity (HC) and extra-high-conductivity (EHC) wires produced from AA6201 alloy, artificial aging is typically performed at 175 °C for more than 6 h or at 195 °C for 4 h to achieve electrical conductivity above the nominal value of 52.5–53% IACS (Ref 17). While these conditions successfully enhance conductivity, they often reduce tensile strength due to the precipitation of incoherent phases. To mitigate this adverse effect, a modified processing route has been proposed to produce HC and EHC wires from AA6201 redrawn rods. This approach minimizes excessive aging exposure and preserves tensile strength at near-maximum levels through low-temperature heat treatment (Ref 17).

In the present study, horizontal solidification experiments were carried out under a transient heat extraction regime using as-cast AA6201 samples. The influence of the solidification thermal parameters, namely growth rate (V_L) and cooling rate (T_R), on microstructural evolution and hardness was subsequently investigated, both before and after T6 heat treatment. Furthermore, the interplay between the characteristic as-cast microstructure and precipitation hardening was systematically evaluated in the context of cold-rolled conductive wire fabrication. The results demonstrate that controlling solidification kinetics combined with optimized T6 treatment enables precise microstructural refinement and uniform distribution of strengthening precipitates, leading to a significant enhancement of both hardness and tensile strength. When applied to cold working, this approach provides an effective strategy to achieve an optimal balance between mechanical strength and electrical conductivity, representing a promising route to produce high-performance AA6201 conductive wires. These findings offer a practical framework for industrial processes aiming to simultaneously maximize conductivity and mechanical performance in Al-Mg-Si alloys.

2. Experimental Procedure

A horizontal directional solidification experiment was carried out using an Al-0.6 Mg-0.8Si (wt.%) alloy in a water-cooled mold, as schematically illustrated in Fig. 1(a) and (b). Temperature measurements were performed along the length of the ingot, and the recorded thermal data were subsequently used to determine the solidification parameters, namely the growth rate (V_L) and cooling rate (T_R), following the methodology depicted in Fig. 1(c).

The alloy was produced with a total mass of 1800 g, which was sufficient to evaluate the composition inside the crucible via thermal analysis based on its *liquidus* and *solidus* temperatures (T_L and T_S , or T_E), before pouring the molten metal into the horizontal ingot mold (Fig. 1b). Using the predicted initial mass, a stoichiometric calculation was performed to determine the exact mass of each constituent element. Magnesium was supplied as a Mg-7Al master alloy (wt.%). The elements

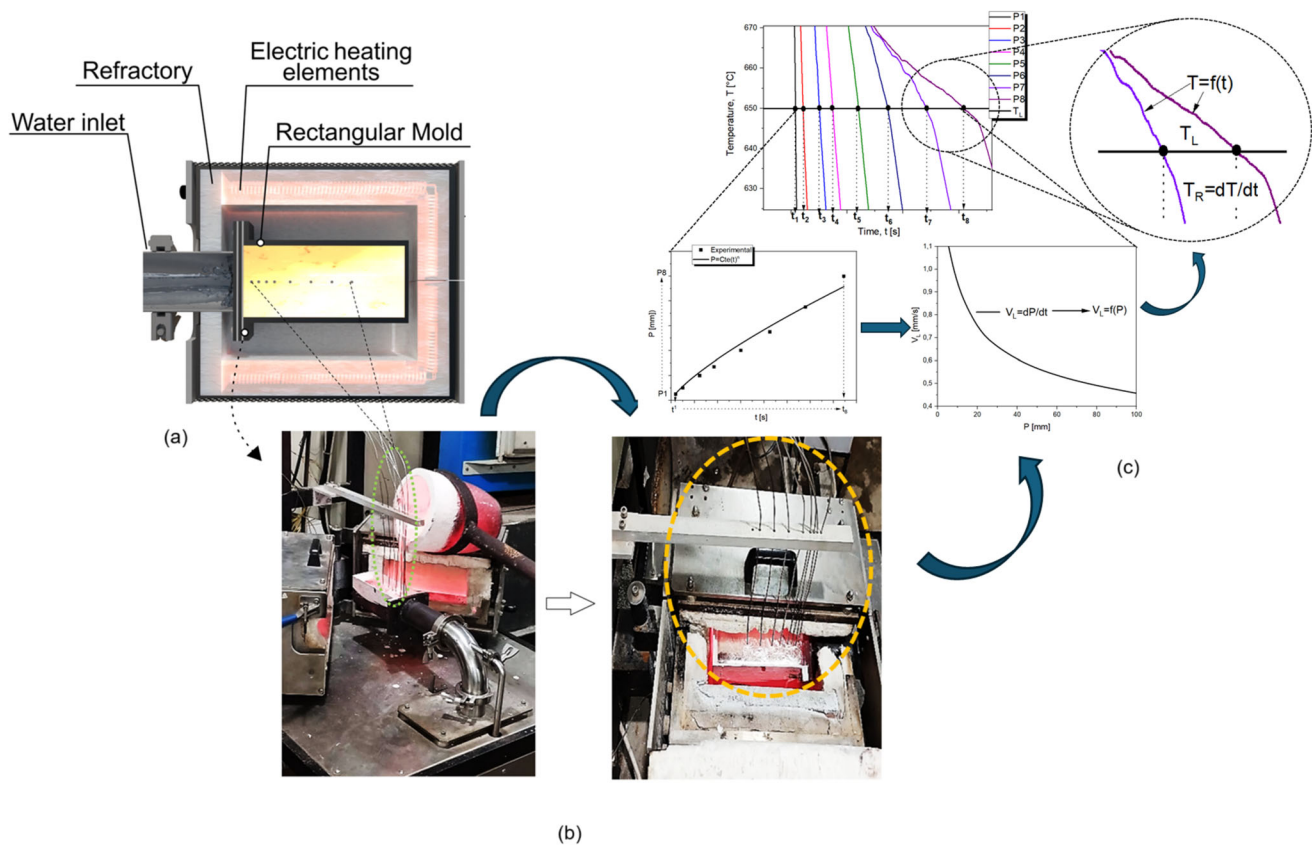


Fig. 1 (a) Schematic of the horizontal solidification setup; (b) schematic of metal pouring and thermocouple placement; (c) detail of the method used for experimental determination of growth rate (V_L) and cooling rate (T_R)

Table 1 Experimental conditions adopted for the production of the investigated Al-Mg-Si alloy

Element	Mass, g	(wt.%)
AA6201 aluminum alloy	1800	100
Si	14.4	0.8
Mg	10.8	0.6
Al	1774.8	98.6

corresponding to the nominal alloy composition were cut and weighed with high precision, as detailed in Table 1. Table 2 presents the chemical compositions of the Al and Si raw materials, as well as the Mg-Al master alloy used for alloy preparation.

To determine the chemical composition of the produced alloy, a thermocouple connected to a temperature recorder was immersed in the molten metal to monitor its natural cooling to room temperature within the crucible. This experimental approach enabled the acquisition of the corresponding cooling curve, which was subsequently used to compare the measured *liquidus* (T_L) and *solidus* (T_S) temperatures with those predicted by the phase diagrams simulated using the Thermo-Calc software. A similar procedure was successfully applied in a recent study to verify the composition of Sn-Ni alloys (Ref 15).

Once the *liquidus* temperature (T_L) of the alloy was confirmed, the crucible containing the molten metal was returned to the muffle furnace for complete melting and subsequently poured into the horizontal solidification device, as

illustrated in Fig. 1(b). The solidified ingot was then demolded and prepared for structural characterization, as shown in Fig. 2(a). The ingot was cut longitudinally, with one section selected for metallographic analysis. This section was sequentially ground using 80-1200 mesh sandpaper to reveal the macrostructure produced by the horizontal solidification process. Poulton's reagent (60 ml of HCl, 30 ml of HNO₃, 5 ml pf HF, and 5 ml of H₂O) was employed for macrostructural etching.

After macrostructural characterization, samples were longitudinally sectioned at positions 5, 10, 15, 19, 30, 50, 70, 90, and 110 mm from the cooled base of the ingot. These sections were sequentially ground (80-1200 mesh) and etched with 5% NaOH to reveal the as-cast microstructure. Secondary dendrite arm spacing (λ_2) was measured following the method proposed by McCartney and Hunt (Ref 22), which involves averaging the distances between adjacent secondary arms on the longitudinal section of selected dendrites, as illustrated in Fig. 2(b).

For the T6 heat treatment (HT-T6), three samples were extracted from the transverse direction of the solidified ingot, i.e., perpendicular to the heat flow direction, at positions 10-20, 40-50, and 90-100 mm from the heat transfer surface. Each specimen measured 10 × 10 × 50 mm. The samples were subjected to heat treatment under the following conditions: solution treatment at 530 °C for 3 h, water quenching at room temperature, and artificial aging at 200 °C for 2.5 h. It is worth noting that the HT-T6 stages—homogenization, quenching, and aging—typically occur at temperatures above 500 °C, in controlled warm water, and between 100-200 °C, respectively. Similar parameters, including temperature and time ranges,

Table 2 Chemical composition (wt.%) of the raw materials used in the preparation of the investigated Al-Mg-Si alloy (Ref 16, 33)

Metais	Al	Ti	Ca	Fe	Ni	Cu	Mg	Si	Cr	Zn
Al	Balance	0.009	...	0.175	0.0148	0.024	0.0011	0.103	...	...
Si	0.1094	0.0455	0.0214	0.3164	0.0102	...	...	Balance	0.27	...
Mg-Al	6.981	...	...	0.001	0.001	0.115	Balance	0.246	...	0.631

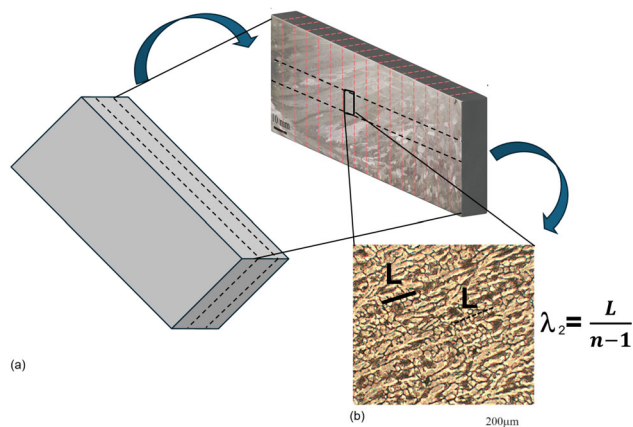


Fig. 2 (a) Illustration of the horizontal ingot sectioning; (b) representation of the procedure used to measure secondary dendrite arm spacing (λ_2)

have been reported in the literature (Ref 9, 13, 14, 17, 23, 24-26) for Al-Mg-Si alloys. Additionally, comparable heat treatment procedures are often integrated into continuous casting and rolling operations, where the alloy undergoes in-line heat treatment during processing.

Rockwell F hardness (HRF) tests were performed on both as-cast and T6-treated samples. Three measurements were taken on each sample surface, approximately at the same locations where the secondary dendrite arm spacings were evaluated. Additionally, since the spherical indenter produces localized plastic deformation zones that may influence subsequent readings, each indentation was performed in regions sufficiently distant from one another at least three times the diameter of the impression. All tests were carried out in strict accordance with the ASTM E18-15 standard to ensure measurement accuracy and reproducibility.

To evaluate the effects of the solidification and heat treatment conditions, two samples one as-solidified and the other T6-treated were machined to a diameter of 9.5 mm, both taken from the same ingot region (90-100 mm). These specimens were subsequently subjected to a cold-rolling process using a MENAC reversible duo mill equipped with interchangeable circular rolls. A total cold work reduction of 68.4% was applied, corresponding to a diameter reduction from 9.5 to 3 mm, achieved through twelve consecutive passes with intermediate reductions. The resulting wire was then tested for electrical and mechanical performance. Electrical resistivity measurements were carried out using a MEGABRÁS MPK-2000 multiohmmeter (Kelvin bridge) in accordance with NBR 5118 (Ref 27, 28) and NBR 6814 (Ref 29, 30) standards, while tensile tests were performed following the NBR 6810 (Ref 31, 32) specifications for electrical conductors.

3. Results and Discussion

3.1 Thermal and Microstructural Analysis

Figure 3 presents the theoretical and experimental cooling curves of the produced Al-0.6 Mg-0.8Si (wt.%) alloy. A good agreement was observed between the experimentally determined *liquidus* (T_L) and *solidus* (T_S) transformation temperatures and those predicted by Thermo-Calc simulations, as shown in Fig. 3a and b, respectively. The experimentally obtained values of $T_L = 650$ °C and $T_E = 557$ °C were therefore adopted. This close correspondence validates the experimental procedure and confirms the compositional accuracy of the prepared alloy.

Figure 4 and 5 presents the thermal data and derived solidification parameters recorded during the horizontal solidification process. Temperature profiles from eight thermocouples embedded in the molten metal are shown in Fig. 4a and were used to determine the solidification kinetics, expressed as the variation of the *liquidus* isotherm position over time ($P = f(t)$, Fig. 4b). The local solidification time (t_{SL} , Fig. 4c) corresponds to the interval between the passage of the *liquidus* (T_L) and *solidus* (T_E) isotherms at the same thermocouple. The $P = f(t)$ relationship shown in Fig. 4b was used to calculate the growth rate (V_L). In turn, the temperature profiles presented in Fig. 4a were employed to determine the cooling rate (TR), as illustrated in Fig. 5a and b.

As shown in Fig. 5, both V_L and T_R decrease along the length of the as-cast ingot, from the cooled end toward the opposite end, reflecting the progressive formation of the solid layer during horizontal solidification. This decrease is also influenced by the thermal resistance at the metal/mold interface, associated with the interfacial “gap” and Newtonian heat transfer effects (Ref 4, 16, 33). These results highlight the strong dependence of local solidification kinetics on heat extraction conditions and ingot geometry, which are critical for controlling the resulting microstructure and mechanical properties of the Al-6201 alloy.

Figure 6 shows the typical solidification structures of the investigated alloy at both macro- and microstructural scales, obtained under the specified processing conditions. The heat flow-related thermal parameters had a decisive influence on the macrostructural length scale. Columnar grains were observed at higher growth rate and cooling rate values, while equiaxed grains predominated under lower V_L and T_R conditions. This behavior clearly defines a columnar-to-equiaxed transition (CET), as indicated by the dashed line in the as-cast ingot region. The columnar-to-equiaxed transition (CET) is a key solidification phenomenon, as it directly affects grain morphology, mechanical isotropy, and defect susceptibility in cast alloys. Although a detailed CET modeling is beyond the scope of the present work, its occurrence is relevant because it reflects

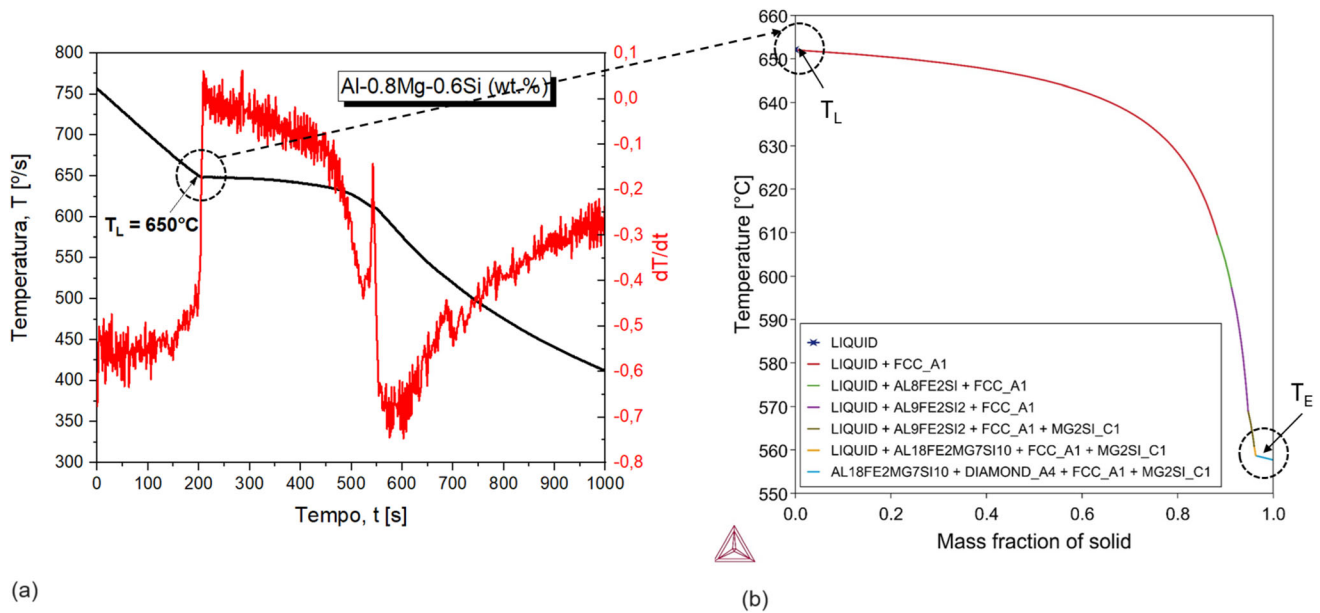


Fig. 3 Solidification paths of the Al-0.6 Mg-0.8Si alloy: (a) experimentally recorded cooling curve; (b) simulated phase transformation path obtained using Thermo-Calc (Source: TCAL8)

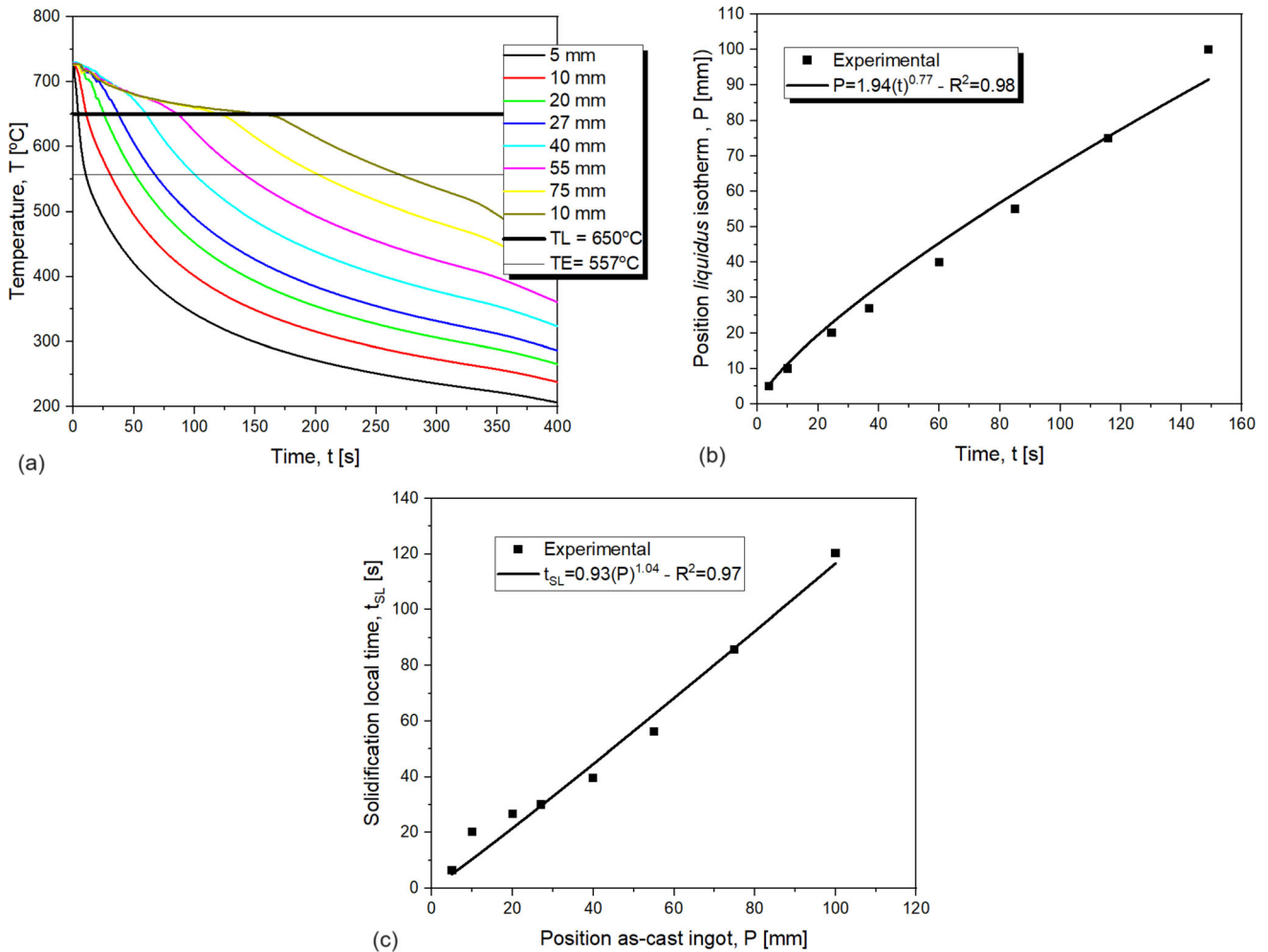


Fig. 4 Thermal data recorded during horizontal solidification of Al-0.6 Mg-0.8Si alloy: (a) temperature profiles from eight embedded thermocouples; (b) variation of the *liquidus* isotherm position with time ($P = f(t)$); (c) local solidification time

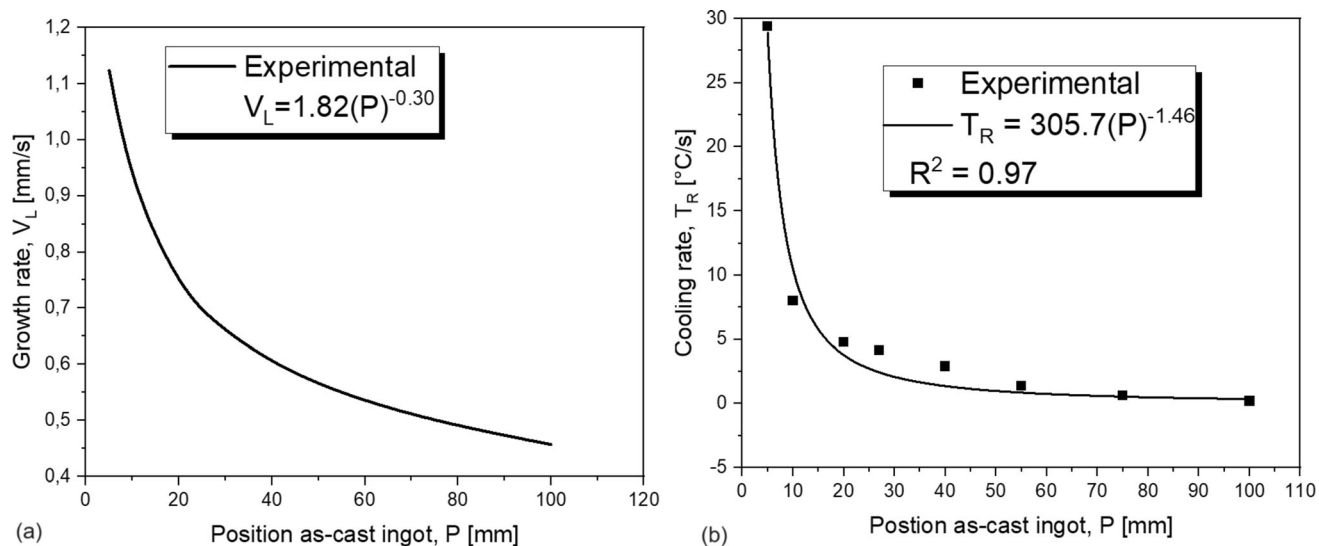


Fig. 5 Transient solidification parameters along the ingot length: (a) growth rate (V_L) and (b) as cooling rate (T_R) as a functions of position, respectively

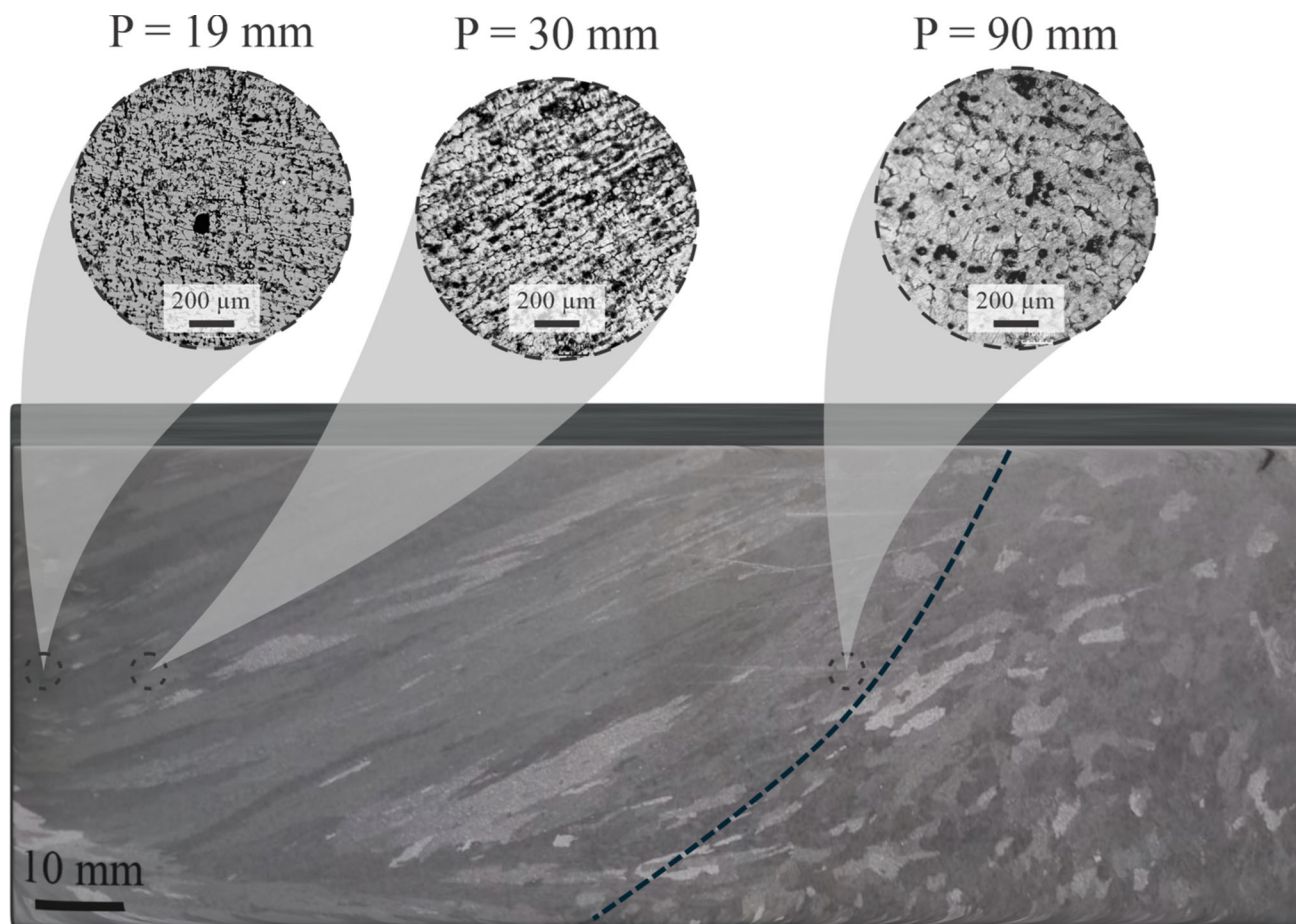


Fig. 6 Typical solidification structures at macrostructural and microstructural scales

the strong influence of thermal parameters on macrostructural evolution, which ultimately impacts mechanical behavior. Although CET analysis is not the main focus of this study, it is well established that this transition strongly affects the mechanical properties of solidified alloys. Therefore, under-

standing the influence of thermal parameters on CET is essential for controlling the microstructural evolution during directional solidification, as also discussed in Reference (Ref 34, 35).

Figure 6 also presents the microstructures obtained at three different positions along the ingot, while Fig. 7 shows the variation in the microstructural length scale, expressed by the secondary dendrite arm spacing (λ_2), as a function of V_L , T_R , and t_{SL} . As observed in Fig. 6, at positions 19, 30, and 90 mm from the heat transfer surface (HTS), the initial solidification conditions strongly influenced the resulting microstructure. Higher V_L and T_R values led to a considerably finer dendritic structure (lower λ_2) near the HTS, whereas coarser structures were observed at positions farther away. This clearly demonstrates that the dendritic spacing is directly affected by the gradual decrease in V_L and T_R during solidification, which results from the increasing thermal resistance associated with the conduction-dominated heat transfer in the growing solid metal. Based on these experimental findings, mathematical relationships were proposed to describe the dependence of λ_2 on V_L , T_R , and t_{SL} , as shown in Fig. 7.

The exponents $-2/3$ and $1/3$ obtained in the derived growth laws, $\lambda_2 = \text{Constant} \cdot (V_L)^{-2/3}$ and $\lambda_2 = \text{Constant} \cdot (t_{SL})^{1/3}$, are in excellent agreement with the theoretical

models proposed by Bouchard and Kirkaldy (Ref 36) and Rappaz and Boettinger (Ref 37). Moreover, these results are consistent with the recent modifications introduced by Ferreira et al. (Ref 38) for thermodynamic nonequilibrium conditions. The $1/3$ exponent also agrees well with previous experimental observations for Al-Si-Mg alloys (Ref 3, 16).

3.2 Mechanical Analysis

Figure 8 presents the results of the F-scale Rockwell hardness (RHF) as a function of the position along the ingot (P), T_R , and λ_2 . The dispersion of the experimental data prevented the establishment of a consistent mathematical correlation between RHF and the analyzed parameters. However, the highest RHF values were observed at positions closer to the heat extraction surface, as shown in Fig. 8a, where higher T_R and lower λ_2 values were also measured (Fig. 8b and c, respectively).

Microstructural refinement is one of the primary mechanisms responsible for increasing mechanical strength in aluminum alloys. Finer dendritic or grain structures enhance

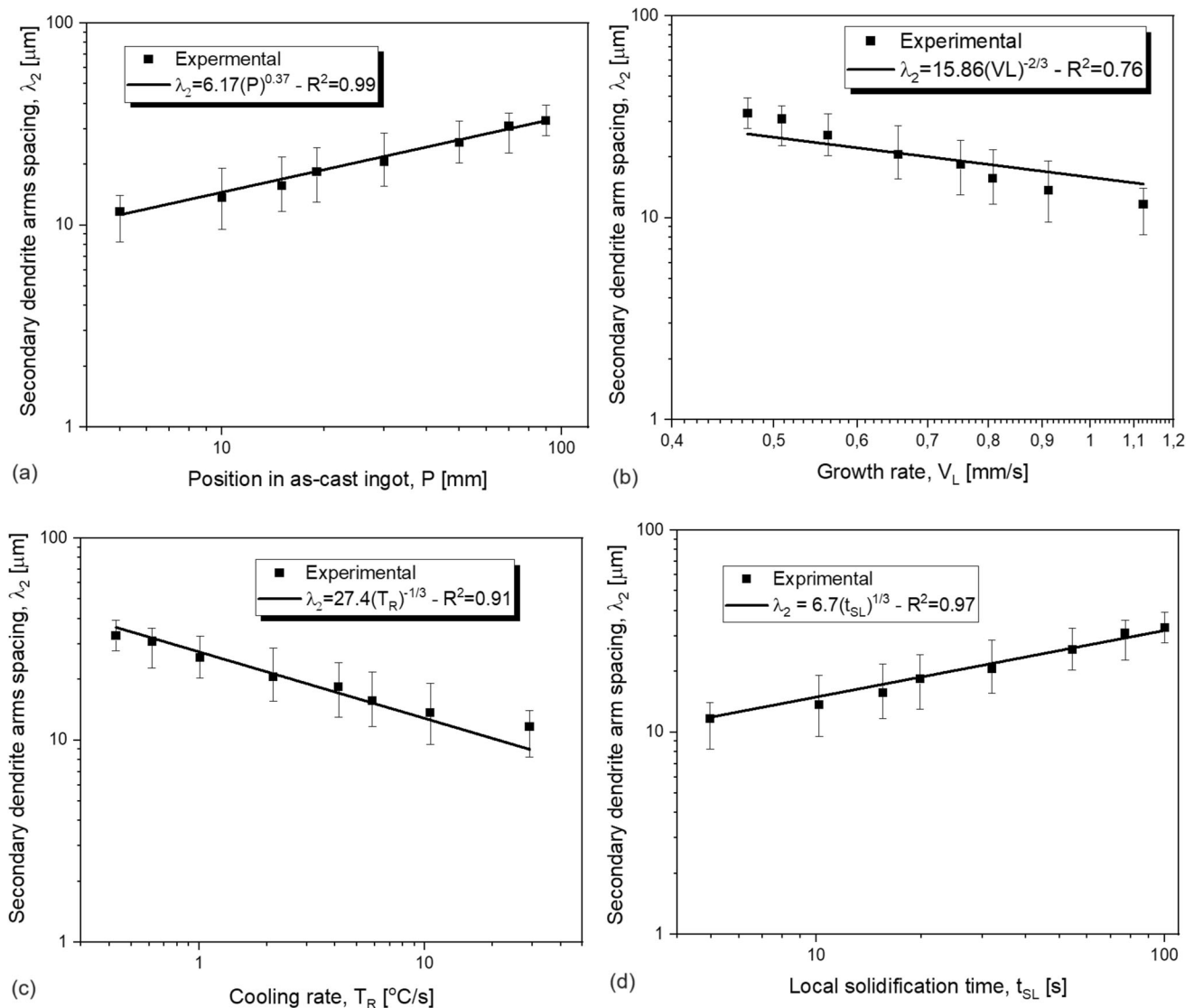


Fig. 7 Dependence of the secondary dendrite arm spacing (λ_2) on: (a) position along the as-solidified ingot, (b) growth rate, (c) cooling rate, and (d) local solidification time

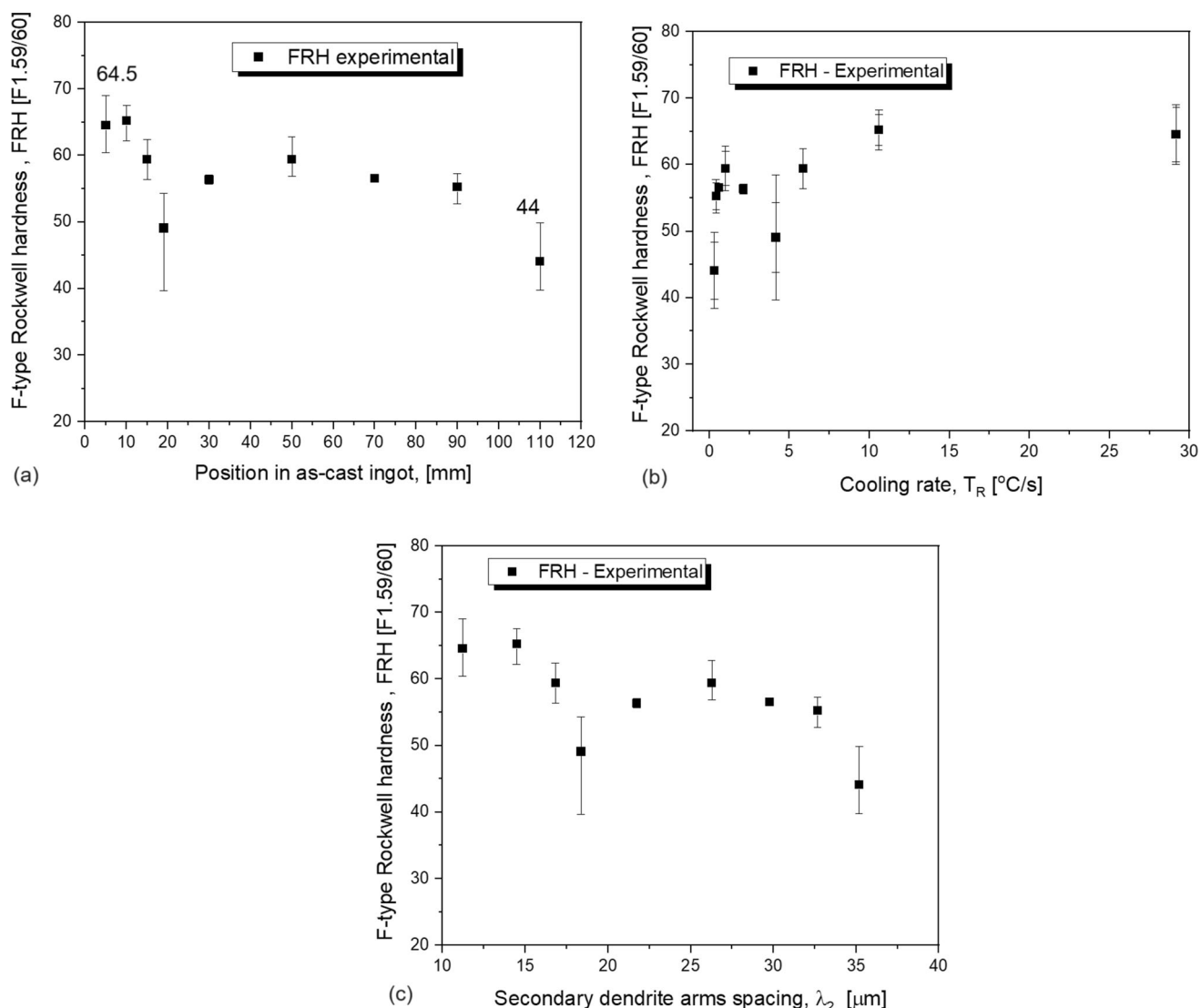


Fig. 8 Experimental Rockwell hardness (HRF) measurements as a function of: (a) position along the as-solidified ingot, (b) cooling rate (T_R), and (c) secondary dendrite arm spacing (λ_2)

hardness because the grain boundaries, and dendrite arms act as barriers to dislocation motion within the solidified alloy, thereby restricting plastic deformation and improving strength.

Figure 9 presents the results of the T6 heat treatment performed under the adopted conditions for three samples. The average hardness increased significantly from 57.6 HRF in the as-cast condition to 87 HRF after heat treatment, corresponding to an improvement of approximately 51%.

It is well established that the addition of magnesium to Al-Si alloys enables precipitation hardening, as Mg reacts with Si to form the β -Mg₂Si intermetallic phase. This phase is primarily responsible for strengthening during the sequential precipitation stages GP $\rightarrow$ β' $\rightarrow$ β'' $\rightarrow$ β , which contribute to the enhancement of the mechanical properties of these alloys (Ref 4, 9).

To evaluate the combined effects of solidification and T6 heat treatment conditions on plastic deformation capacity, two samples, one as-cast and one heat-treated, were selected from the same position along the ingot (90-100 mm) and subjected to a cold-rolling test, as shown in Figure 10. The hardening level achieved by the T6 heat treatment completely suppressed

the plastic deformability of the heat-treated sample, making cold rolling unfeasible under the adopted conditions. In contrast, the as-cast sample successfully underwent the forming process, exhibiting high plastic deformation capacity, as evidenced by the 3 mm diameter wire obtained after cold rolling (Fig. 10).

The β -Mg₂Si phase, which precipitates within the α -Al matrix during heat treatment rather than remaining confined to the interdendritic regions as in the as-cast condition, appears to act as a more effective barrier to dislocation motion, thereby reducing the alloy's plastic deformability.

The conductive wire obtained through cold rolling was subjected to tensile and electrical tests under the adopted conditions, and the results are presented in Fig. 11. The measured ultimate tensile strength, electrical resistivity, and conductivity of the cold-rolled wire were 320.48 MPa, 33.6 nΩ·m, and 51.2% IACS, respectively. These results demonstrate that for the assumed solidification conditions characterized by an average secondary dendritic arm spacing (λ_2) of approximately 33 μm at the as-cast position selected for rolling, and considering the degree of cold work achieved through three

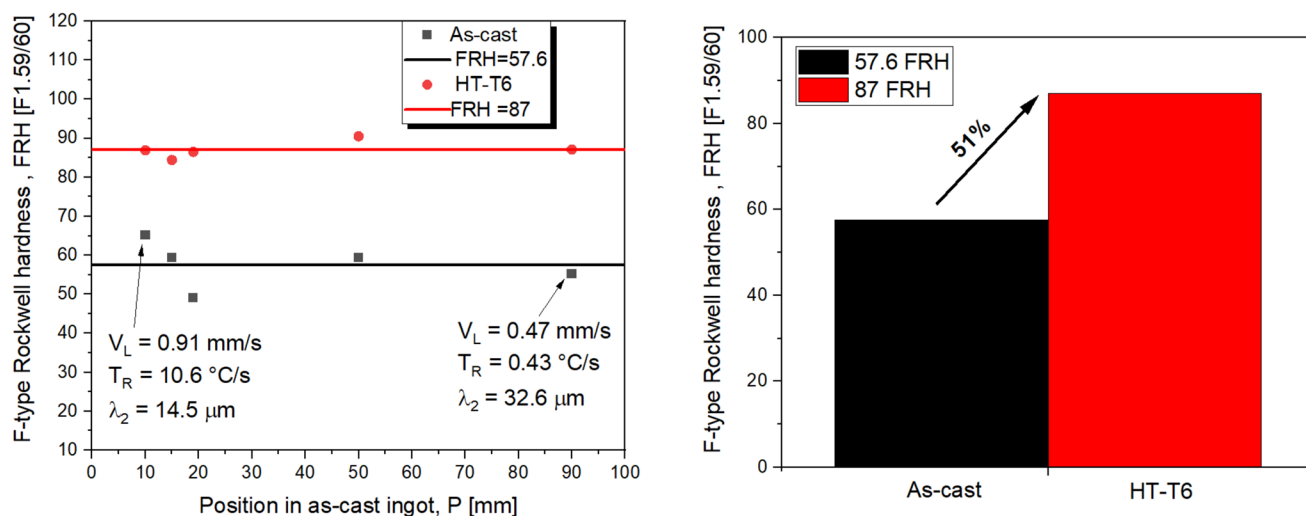


Fig. 9 Comparison of Rockwell hardness (HRF) between the as-cast and T6 heat-treated samples. The T6 condition exhibited a significant hardness increase from 57.6 HRF (as-cast) to 87 HRF (T6), corresponding to an improvement of approximately 51% due to precipitation hardening by the β -Mg₂Si phase

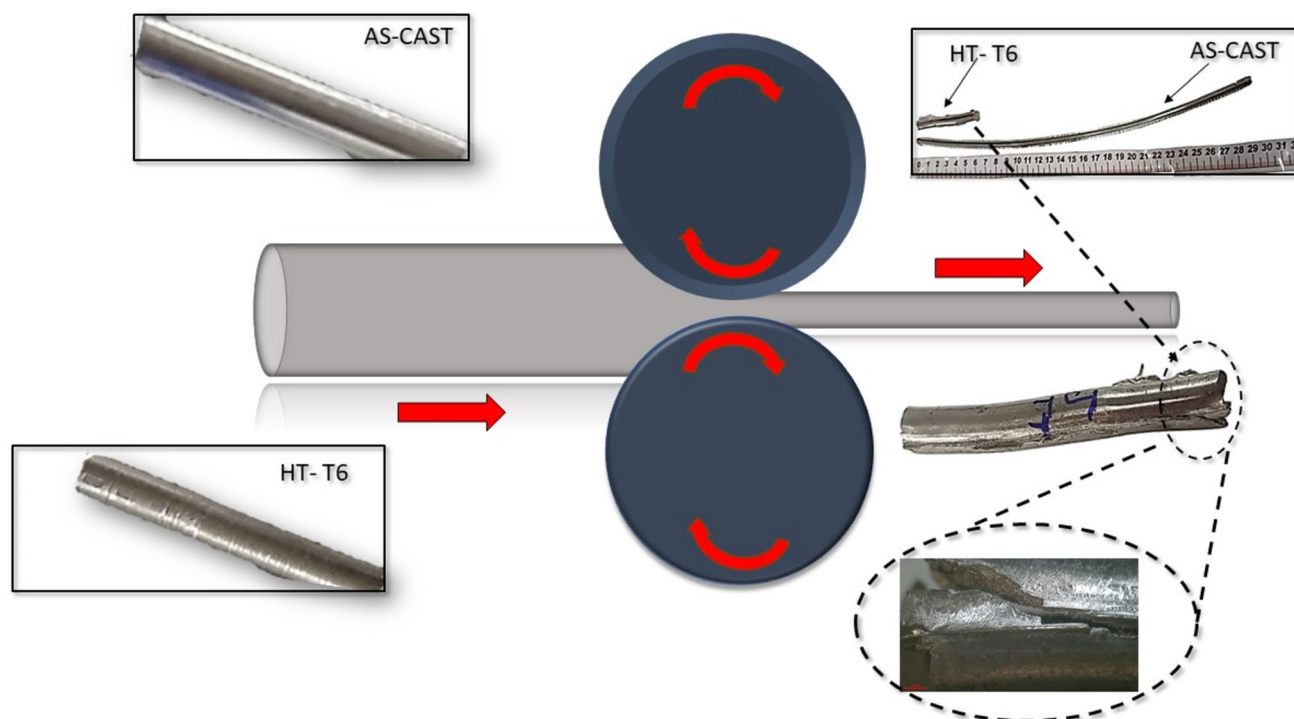


Fig. 10 Schematic representation of the results after cold rolling of the as-cast and T6 heat-treated (HT-T6) sample

rolling passes to produce the 3 mm wire, the material exhibited an excellent balance between mechanical strength and electrical conductivity.

Compared with previous investigations, the present conductivity level (51.2% IACS) is consistent with the findings reported by Khangholi et al. (Ref 13). The authors observed that Al-xSi-0.65 wt% Mg alloys ($x = 0.4-0.89$) exhibited conductivities below the minimum required value of 52.5% IACS at the peak-aged condition. Smyrak et al. (Ref 39)

reported that optimal aging parameters (150 °C for 4-10 h) in Al-Mg-Si wire rods yield tensile strength and resistivity values of 200 MPa and 30 nΩ·m, respectively, which are ideal for achieving a high strength-to-conductivity ratio. In contrast, the investigated alloy (upward solidified Al-6201) exhibits a significantly higher tensile strength but slightly lower conductivity, illustrating the typical strength–conductivity trade-off observed in age-hardened aluminum alloys.

Moreover, the achieved combination of properties is comparable to those obtained for high-conductivity (HC) and extra-high-conductivity (EHC) wires produced from AlB₂-modified AA6201 alloys, which display improved mechanical strength without compromising conductivity (Ref 17).

In order to compare the hardening mechanisms applied in the present work—namely, the T6 heat treatment and cold working, the average hardness values presented in Fig. 9 were converted to Brinell hardness (BH) using the conversion table provided in the ASTM E40-12b^{e1} standard. This conversion was carried out by applying a third-degree polynomial interpolation to the data indicated in the standard, given by the mathematical expression: $BH = 97.62 + 6.61(HRF) - 0.10(HRF)^2 + 5.96 \times 10^{-4}(HRF)^3$. The calculated BH values were subsequently converted into ultimate tensile strength (σ_{UTS}) using the relationship $\sigma_{UTS} = \alpha \cdot BH$, where α was assumed to be equal to 4 for aluminum alloys, as recommended

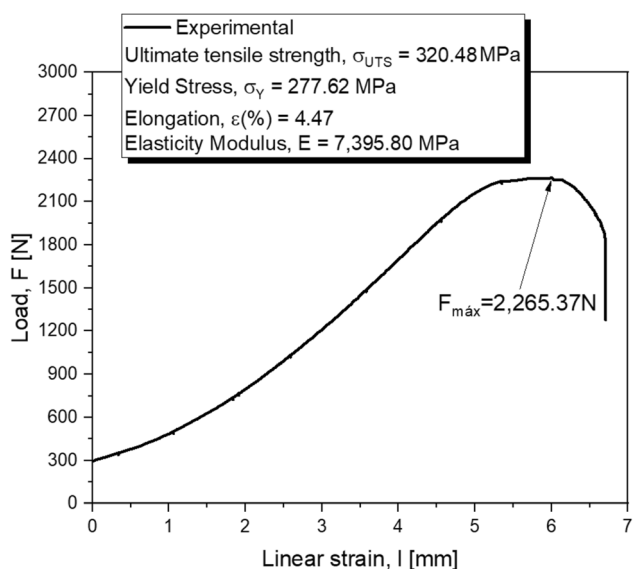
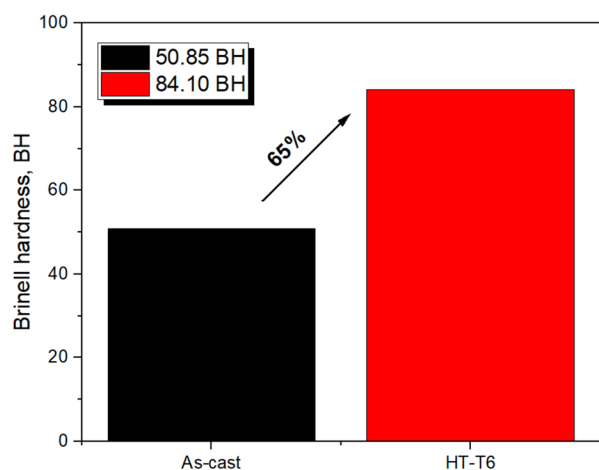


Fig. 11 Experimental load–strain curve of cold-rolled wire of the investigated alloy



in the literature (Ref 38). The results, consolidated in Fig. 12, clearly demonstrate that the calculated σ_{UTS} value (336.4 MPa) is higher than that obtained after cold working (320.48 MPa), indicating that plastic deformation by cold rolling was not viable for the heat-treated sample, as depicted in Fig. 10.

In order to confirm the phase transformations indicated in the solidification path shown in Figure 3b, scanning electron microscopy (SEM) analyses and energy-dispersive spectroscopy (EDS) microanalyses were performed, and the results are presented in Figures 13 and 14. The SEM/EDS micrographs in Figure 13 were obtained from samples etched with 5% NaOH, whereas those in Fig. 14 correspond to polished samples without chemical etching.

The EDS elemental maps of the as-cast sample (Fig. 13a) reveal the presence of Si and Mg concentrated in the interdendritic regions, suggesting the formation of Si and Mg₂Si intermetallic phases, while Fe was found to be uniformly distributed throughout the structure. After the T6 heat treatment (Fig. 13b), the EDS maps indicate that Mg became more homogeneously distributed within the Al-rich (Al- α) matrix, confirming its dissolution and subsequent retention as a supersaturated Mg₂Si phase precipitated within the matrix. In contrast, Si and Fe remained segregated in the interdendritic channels, as expected, since both elements are known to exhibit negligible solubility during the solution treatment stage.

The Fe-rich intermetallic phase was clearly observed within the interdendritic regions in the unetched micrographs of Fig. 14. Therefore, although x-ray diffraction (XRD) analysis was not performed in this study, the SEM/EDS results confirm that the typical solidification microstructure of the investigated alloy consists of an Al- α matrix, Si, Mg₂Si, and AlSiMgFe-type intermetallic compounds, consistent with the solidification path proposed in Fig. 3b and with typical microstructures reported in the literature for Al-Si-Mg alloys (Ref 3-5, 40-42).

4. Conclusions

1. The growth and cooling rates (V_L and T_R) imposed by the horizontal water-cooled solidification system promoted the formation of columnar and equiaxed grains under high and low T_R and V_L conditions, respectively, characterizing a columnar-to-equiaxed transition (CET).

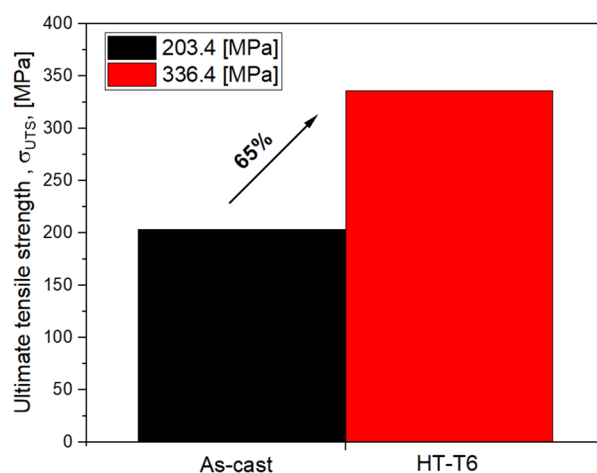


Fig. 12 Converted hardness and ultimate tensile strength (UTS) values obtained according to ASTM E40- for Peer Review 12b&1

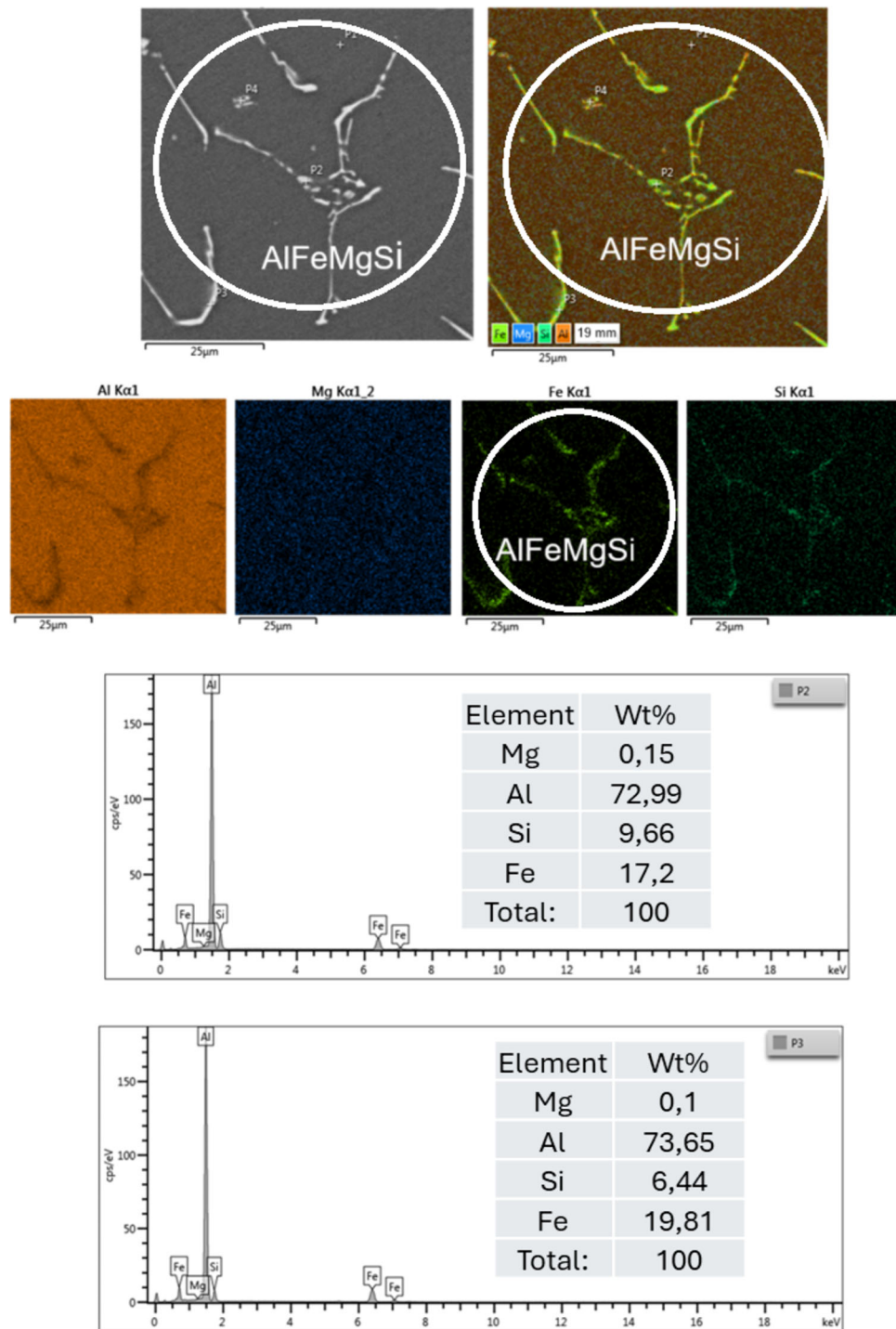


Fig. 14 SEM micrographs and EDS elemental microanalysis of the T6 heat-treated sample without chemical etching

This provides valuable guidance for metallurgical processing industries to control the solidification thermal parameters to tailor the grain structure for specific applications.

2. The growth and cooling rate levels achieved during directional solidification resulted in a typical dendritic network along the length of the as-cast ingot.
3. Experimental relationships of the types $\lambda_2 = \text{Constant} \cdot V_L^{-2/3}$, $\lambda_2 = \text{Constant} \cdot T_R^{-1/3}$, $\lambda_2 = \text{Constant} \cdot t_{SL}^{1/3}$ effectively described the dependence of the secondary dendrite

arm spacing (λ_2) on the solidification thermal parameters. These results indicate finer microstructures for ingot regions closer to the heat extraction surface, where higher V_L and T_R values prevailed.

4. The strengthening mechanism associated with finer microstructures, acting as barriers to dislocation motion, was confirmed, as higher hardness values were measured for lower λ_2 values.
5. A precipitation-hardening mechanism resulting from the T6 heat treatment applied to the as-cast samples was also

evidenced, producing a 51% increase in hardness and a corresponding 65% increase in ultimate tensile strength.

6. The typical solidification dendritic microstructure ($\lambda_2 = 32.6\text{--}33.9\ \mu\text{m}$) enabled the production of 3 mm conductive wire through successive cold rolling passes. This processing route was not feasible for the T6-treated samples due to excessive hardening caused by precipitation.
7. Considering the achieved results for ultimate tensile strength (320.48 MPa), electrical resistivity (33.6 n $\Omega\cdot\text{m}$), and electrical conductivity (51.2% IACS), the combination of directional solidification and cold rolling under the conditions adopted in the present study emerges as a promising route for manufacturing conductive wires from 6201 aluminum alloy.

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Conflict of interest

None.

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